

Lab Exercise- 4

Submission and Demo Date on or before **05.10.2024**

(It is mandatory to show demo today for one component atleast)

1. Design Enterprise network design using CISCO Packet Tracer:

Sample Scenario:

The three departments, namely, Marketing, Sales and General Support, should have separate LAN with individual applications. However, these departments must share data.

Location and number of workstations

Marketing department occupying the first floor with 20 workstations.

Sales department on the second floor with 30 workstations.

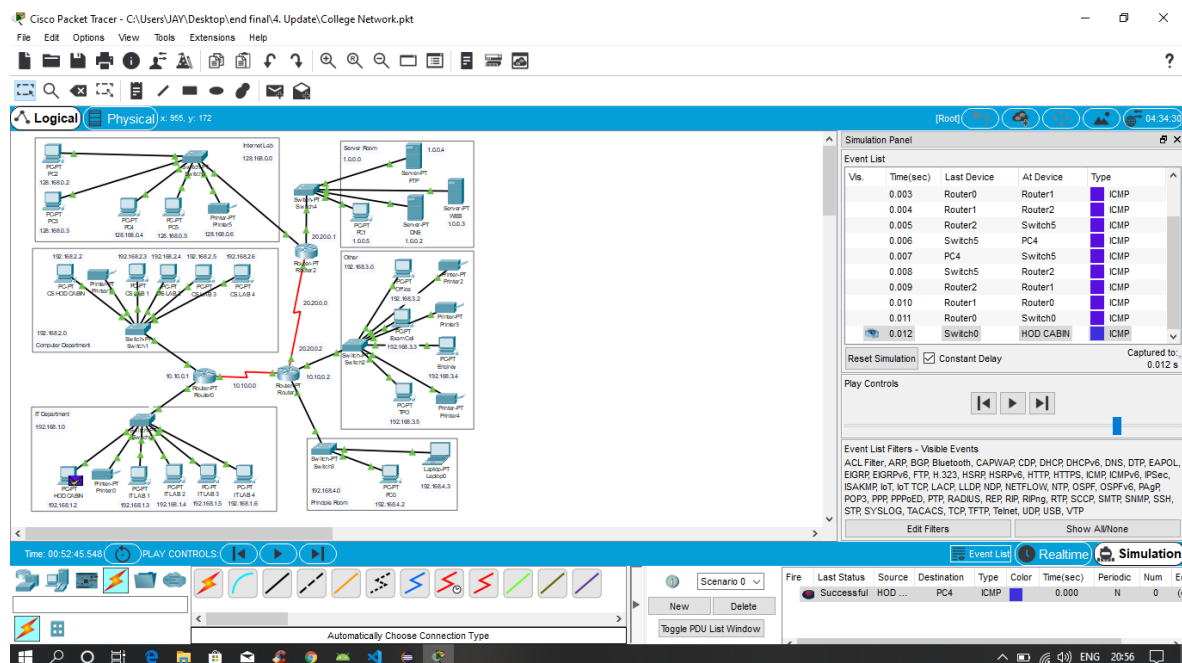
General support with 25 workstations on the last floor

Access to internet, intranet, and email. Internet connectivity should be configured for all users accessing the network.

(OR)

2. Design the campus network using Packet Tracer.

Sample:



<https://github.com/katejay/College-Network>

3. Implement a program to allocate the IP addresses to the Enterprise for the sample Scenario Given:

An ISP is granted a block of addresses starting with 190.100.0.0/16 (65,536 addresses). The ISP needs to distribute these addresses to three groups of customers as follows:

- The first group has 64 customers; each needs 256 addresses.
- The second group has 128 customers; each needs 128 addresses.
- The third group has 128 customers; each needs 64 addresses.

Find Starting address and end address of each group. How many addresses are still not allotted to any groups.

4. **Implement Link State Routing Algorithm/Distance Vector Routing** for the topology given and construct routing a=table for any node:

